

⑦ Improper Integrals. ⑧

1. The definition of Type 1.

a). $\int_a^t f(x) dx$ exists for every number $t \geq a$, then

①. 区间 $\rightarrow \infty$ / $-\infty$ 时不存在.

②. 区间 t 在区间 $[a, b]$ 上不连续 / 不满足黎曼可积的定义:

b). $\int_t^b f(x) dx$ exists for every number $t \leq b$, then

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx.$$

Riemann's Original Definition of the Integral: A bounded function f is integrable on $[a, b]$ with $\int_a^b f = A$ if for all $\epsilon > 0$ there exists a $\delta > 0$ such that for any tagged partition $(P, \{c_k\})$ satisfying $\Delta x_k < \delta$ for all k , it follows that $|R(f, P) - A| < \epsilon$.

provided the limit exists (a finite number).

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx.$$

$\Rightarrow$ For a). b). $\int_a^\infty f(x) dx$ and $\int_{-\infty}^b f(x) dx$ are called convergent if the corresponding limit exists. divergent $\Rightarrow$ does not exist.

c). If both $\int_a^\infty f(x) dx$ and $\int_{-\infty}^a f(x) dx$ are convergent $\Rightarrow \int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx \quad \forall a \in \mathbb{R}.$

Δ The important improper integral:

$$\int_1^\infty \frac{1}{x^p} dx \Rightarrow \lim_{t \rightarrow \infty} \int_1^t x^{-p} dx = \lim_{t \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_{x=1}^{x=t} = \lim_{t \rightarrow \infty} \frac{1}{1-p} \left[\frac{1}{t^{p-1}} - 1 \right]$$

① if $p > 1$, then $p-1 > 0$. So as $t \rightarrow \infty$, $t^{p-1} \rightarrow \infty \Rightarrow 1/t^{p-1} \rightarrow 0$.

$$\int_1^\infty \frac{1}{x^p} dx = \frac{1}{p-1} \quad \text{if } p > 1. \Rightarrow \text{converges.}$$

② if $p < 1$, then $p-1 < 0$. $\frac{1}{t^{p-1}} = t^{1-p} \rightarrow \infty$ as $t \rightarrow \infty$. $\Rightarrow$ diverges.

② $\int_1^\infty \frac{1}{x^p} dx$ is convergent if $p > 1$ and divergent if $p \leq 1$.

2. Definition of an improper integral of Type 2.

a). if f is continuous on $[a, b)$ and is discontinuous at b , then.

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx.$$

b). if f is continuous on $(a, b]$ and is discontinuous at a , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx.$$

$\Rightarrow$ as a finite num

$\Rightarrow$ For a). b). $\int_a^\infty f(x) dx$ and $\int_{-\infty}^b f(x) dx$ are called convergent if the corresponding limit exists.
divergent $\Rightarrow$ does not exist.

c). if f has a discontinuity at c , where $a < c < b$. and $\int_a^c f(x) dx$, $\int_c^b f(x) dx$ are convergent.
 $\Rightarrow \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

$\triangle$ 即存在无穷间断点时 (infinite discontinuity).

3. The comparison Test for Improper Integrals

Comparison Theorem Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

- (a) If $\int_a^\infty f(x) dx$ is convergent, then $\int_a^\infty g(x) dx$ is convergent.
- (b) If $\int_a^\infty g(x) dx$ is divergent, then $\int_a^\infty f(x) dx$ is divergent.